

Password saving: Another common unsafe practice in vogue is the way we save passwords. People usually store the passwords in plain text, mostly on sticky notes or a notepad. Some even store their passwords on the browser, which makes it convenient for users but also puts their online accounts in serious jeopardy.

Randomness

Randomness is one of the desired qualities in a password but yet, it is very difficult to produce. Each time you create a password, you typically use bits of data from your personal space to help in remembering the passwords. The problems arise when the number of accounts per user increases. Then the new password that is generated is once again from your personal data, which is hardly random.

So, for example, if you are creating an account for a social media service, you would provide the following details:

1. First name
2. Last name
3. Date of birth
4. Email ID

If you look closely at the above information, this is unique to you, the individual user, and when you create a username or in our case, a password, even that is based on the above data. If an account is set up with a unique password created from personal information, when a new password needs to be set up for another online service, the options are again limited to your basic personal information. So users end up making just a few modifications to the old password, such as adding sequential numbers to it or substituting special characters. This, however, provides very little protection because when only limited modifications are made to the old password, so that users can remember the new modified one, it trumps the randomness required to make passwords secure.

One could try to set up random passwords but they would hardly be random because in order to avoid the risk of forgetting the passwords, users again pick chunks of data which they remember well. Even if they succeed in creating truly random passwords, it would be difficult to keep track of all those random passwords unless you possess a photographic memory.

Password managers

Password managers help create secure passwords and store them in an encrypted file but they too have their limitations, such as:

- (a) Password managers store the passwords in an encrypted file; so if that file is accidentally deleted or corrupted for some unknown reason, there is no way to access those passwords.
- (b) If password managers are simultaneously used in a cell phone and a personal computer, the devices may need to be synced so that the information gets updated regularly.
- (c) If a completely new device that doesn't have the password

manager app is used, accessing the passwords may not be possible.

- (d) Occasionally, the user may need to use the Internet to sync the passwords between various devices.
- (e) Some password managers encourage users to store their passwords in the cloud, which is not controlled by the users, who then run the risk of handing over sensitive information to some business entity that has unlimited access.

The Master Password app

The Master Password app is not a password manager, yet it has a striking resemblance to one. It is a stateless password generator app that produces random and secure passwords for your use.

Master Password uses the following details to create passwords:

- (a) Your name
- (b) Master password
- (c) Site name

When you open the Master Password app, it will look like what is shown in Figure 1.

Name: The name is the first thing you need to enter in order to generate passwords. It is recommended that you use your full name, i.e., your first name, middle name and last name because an intruder will then have to know all the parts of your name to derive your passwords.

Master password: This is the only part which you have to remember. This is also one of the integral parts in the Master Password app. So it is important you set a strong and secure password for this part.

Although we are all aware of the best practices for creating passwords, the following recommendations will add more security to your passwords.

- (a) Try not to include too much personal information that has already been made public.
- (b) Use a brand new password that hasn't been used before.
- (c) If you're using phrases and quotes as your passwords, try to turn them into an acronym.
- (d) Avoid using dictionary words, and even if you have to use them, modify their spelling but let their pronunciation remain intact.

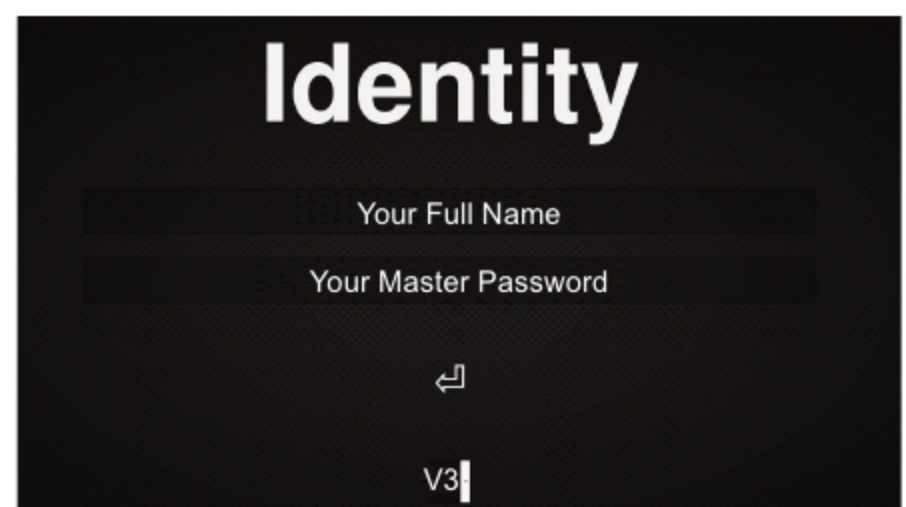


Figure 1: Master Password start page